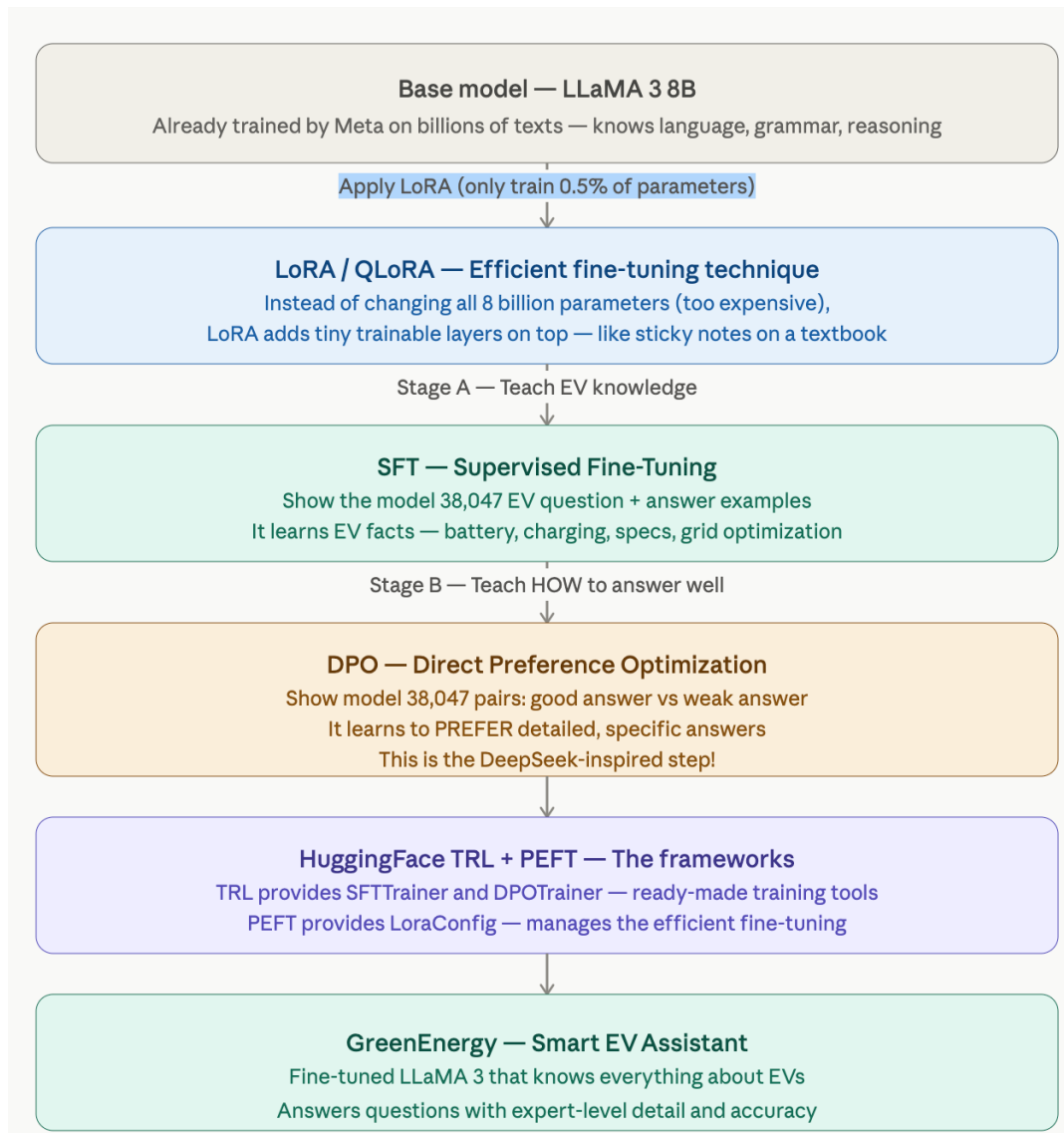


Basic Architecture



Objectives

Objective 1 — Domain Adaptation via SFT

Fine-tune LLaMA 3 8B on 38,047 EV-specific instruction examples using Supervised Fine-Tuning (SFT) with LoRA adapters, teaching the model deep knowledge of:

- Battery technology and health metrics
- Charging infrastructure and speeds
- Vehicle specifications and performance
- Grid optimization and V2G technology
- Cost, sustainability, and market data

Objective 2 — Response Quality via DPO

Apply Direct Preference Optimization (DPO) on 38,047 preference pairs (chosen vs rejected answers) to teach the model to prefer detailed, data-driven, specific responses over vague, generic ones — directly inspired by DeepSeek's reward-based training philosophy.

Objective 3 — Parameter Efficiency

Achieve domain adaptation using LoRA (Low Rank Adaptation) targeting only the attention and feed-forward layers, training less than 1% of total model parameters while preserving the base model's general language capabilities.

Objective 4 — Agentic Deployment

Wrap the fine-tuned model in an agentic architecture with conversational memory and multi-step reasoning, served via a FastAPI REST API — demonstrating the thesis claim that LLM-orchestrated agentic systems improve autonomy and performance on complex domain-specific tasks.

What we have to upload on GPU?

There are different methods but few of them I mentioned below. But I think SSH method is quite easy.

Method 1 — SSH

open bash

```
scp -r ev_agent/training/ user@[server-address]:~/ev_agent/training/
scp -r ev_agent/data/processed/ user@[server-address]:~/ev_agent/data/processed/
scp ev_agent/requirements.txt user@[server-address]:~/ev_agent/
```

Method 2 — Jupyter Notebook

1. Open Jupyter in browser
2. Click Upload button (top right)
3. Upload these files one by one:

```
training/train_config.yaml
training/finetune_sft.py
training/finetune_dpo.py
data/processed/train.json
data/processed/val.json
data/processed/dpo_train.json
data/processed/dpo_val.json
requirements.txt
```

Method 3 — Zip everything and share

Open bash again please

```
cd ev_agent
zip -r greenenergy_training.zip \
```

```
training/\  
data/processed/\  
requirements.txt
```

after choosing the method and uploading the files on gpu server bahi please follow the instruction below.

Now the question is What we have to runs on the server?

Once files(training folder) are uploaded, we have to run these commands in exact order:

Step 1 — Install requirements

```
bash  
pip install -r requirements.txt
```

Step 2 — Verify GPU is working

```
bash  
python3 -c "  
import torch  
print('GPU available:', torch.cuda.is_available())  
print('GPU name:', torch.cuda.get_device_name(0))  
print('GPU memory:', torch.cuda.get_device_properties(0).total_memory / 1e9, 'GB')  
"
```

This should show something like:

```
GPU available: True  
GPU name: NVIDIA A100  
GPU memory: 40.0 GB
```

Step 3 — Run SFT training (Stage A)

```
bash  
cd ev_agent  
python3 training/finetune_sft.py  
Wait 2-4 hours
```

Step 4 — Run DPO training (Stage B)

```
bash  
python3 training/finetune_dpo.py  
Wait 1-2 hours
```

Step 5 — Verify model was saved

```
bash  
ls -lh models/sft_model/  
ls -lh models/dpo_model/
```

Most Important thing about the base model download is to run the file named **finetune_sft.py** and it will download the LLaMA 3 8B from Hugging Face automatically on the server and the size is about 15GB maybe bahi it takes some time So what we need to download the model.

1. A Hugging Face account
2. Accept LLaMA 3 license at: huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct

Below is the account credentials.

Email: mamoonkhan49@gmail.com

password: [Comsats@1994Khan](#)

and please be make sure during step 2 — bahi you needs to accept the license before running the script, otherwise it will fail.
